

ExamiQ+

User Flows, System Locations, and Panel Q&A

Pure logic reference for thesis defense

This document maps each user role to where they act in the system, the step-by-step logic for major functions, and panelist questions with suggested answers.

Part 1 — System Overview

ExamiQ+ is a self-regulated mathematics exam review platform for college students. It evaluates whether confidence-aware review improves calibration awareness and self-regulated learning (SRL). Content is math-only and scoped by degree program. Eight programs exist: Computer Science, Information Technology, BSEd Mathematics, Psychology, Marketing, Human Resources, Hospitality Management, and Criminology.

The system operates in three phases: (1) campus and content setup, (2) student review cycle, and (3) instructional and institutional response. Each phase involves different roles at different locations in the application.

1.1 User Roles and Entry Points

Role	Registration	Approval	Post-Login Home
Student	/accounts/signup/	Immediate (approved, active)	/student/dashboard/
Faculty (Professor)	/accounts/signup/	Pending until campus admin approves	/professor/courses/
Chairperson	/accounts/signup/ (department required)	Pending until campus admin approves	/chairperson/dashboard/
Campus Administrator	Provisioned by admin (not public signup)	Active on creation	/campus/ or /admin/

1.2 Role-Based Access Logic

- Students may only access /student/ routes and shared profile routes.
- Faculty may only access /professor/ routes for courses they own.
- Chairpersons see department-scoped data at /chairperson/ based on their department.
- Campus administrators manage users, sections, and calendar at /campus/.
- Cross-role URL access returns HTTP 403. Unauthenticated access redirects to login.

1.3 Core Modules

Module	Purpose	Primary Actors
User Management	Registration, login, profile, approval	All roles
Academic Content	Sections, assignments, topics, question bank	Campus admin, chairperson, faculty
Exam Module	Timed review, grading, confidence, feedback	Student, faculty
Summary Module	Dashboards, session history, progress	Student, faculty
Reports Module	Analytics, interventions, research export	Faculty, chairperson

Part 2 — Where Users Go for Each Function

2.1 Authentication and Account Management

Function	Who	System Location (URL)	Logic
Register	Student, Faculty, Chairperson	/accounts/signup/	Role selected at signup. Student is active immediately. Faculty and chair enter pending status.
Log in	All	/accounts/login/	On success, redirect to role dashboard. Pending/rejected faculty and chair are blocked.
Log out	All	/accounts/logout/	Session ends; redirect to login.
Edit profile	All	/profile/	Name, photo, email, password. Students also set home degree program, year level, section.
View notifications	All	/profile/notifications/	System notifications for the authenticated user.
Approve registration	Campus admin	/campus/users/ → /campus/users/<pk>/approve/	Activates pending faculty or chairperson accounts.
Reject registration	Campus admin	/campus/users/	Sets rejected status; user cannot log in.
Create faculty/chair account	Campus admin	/campus/users/create/professor/ or /create/chairperson/	Direct provisioning of approved active accounts.

2.2 Campus and Academic Setup

Function	Who	System Location (URL)	Logic
Set current academic year/term	Campus admin	/campus/academic-calendar/	Required gate: students cannot start review without a current term.
Create program sections	Campus admin	/campus/sections/	Defines cohorts with capacity for student signup and assignments.
View campus dashboard	Campus admin	/campus/	Pending registrations and setup status overview.
Create teaching assignment	Chairperson	/chairperson/assignments/create /	Links faculty + section + subject + term. Auto-creates course and exam setup.
Manage teaching assignments	Chairperson	/chairperson/assignments/	Primary way students receive subjects for review.

2.3 Faculty — Course and Content Management

Function	Who	System Location (URL)	Logic
View all courses	Faculty	/professor/courses/	Lists courses owned by the logged-in professor.
Create course	Faculty	/professor/courses/create/	Self-create when no chair assignments exist.
Configure exam setup	Faculty	/professor/courses/<pk>/exam-setup/	Enable review, set topics, difficulties, duration, per-question timer. Gates student eligibility.

Function	Who	System Location (URL)	Logic
Manage topics	Faculty	/professor/courses/<pk>/topics/	Topic list for the course program.
Create questions	Faculty	/professor/courses/<pk>/questions/create/	New questions enter PENDING status until chair approval.
View question bank	Faculty	/professor/courses/<pk>/questions/	List, edit, deactivate questions. Edit of approved question returns to PENDING.
AI-generate questions	Faculty	/professor/courses/<pk>/questions/ai-generate/	Draft questions via LLM; still requires approval.
Edit feedback/explanations	Faculty	/professor/courses/<pk>/feedback/	Improve step-by-step explanations on high-mistake items.
Manage review windows	Faculty	/professor/courses/<pk>/windows/	Optional scheduled review windows (timed or practice mode on model).

2.4 Chairperson — Oversight and Approval

Function	Who	System Location (URL)	Logic
Programs dashboard	Chairperson	/chairperson/dashboard/	Department-scoped per-program performance metrics.
Cross-program analytics	Chairperson	/chairperson/analytics/by-program/	Analytics grouped by student home degree program within department.
Course audit	Chairperson	/chairperson/courses/	Courses with review participation from department students.
Approve/reject questions	Chairperson	Intended: /chairperson/questions/review/	Service layer exists. Approval limited to programs managed by chair's department.
Faculty audit logs	Chairperson	/chairperson/logs/faculty/	Activity logs for faculty in department scope.
Student audit logs	Chairperson	/chairperson/logs/students/	Activity logs for students in department scope.
Research data export	Chairperson	Intended: /chairperson/research/export/	Anonymized CSV with calibration and survey columns for thesis analysis.

2.5 Student — Review and Learning

Function	Who	System Location (URL)	Logic
View performance dashboard	Student	/student/dashboard/	Accuracy, calibration summary, recommendations.
Start review	Student	/student/review/setup/	Wizard: subject → topic → difficulty → duration. Filtered to home program.
Take timed exam	Student	/student/review/<pk>/	Per-question countdown. Confidence inferred from response time.
Submit answer	Student	/student/review/<pk>/answer/<question_id>/	Answer graded; wrong answers logged as mistakes.
View session summary	Student	/student/review/<pk>/summary/	Calibration matrix, accuracy, recommendations. Explanations shown here, not during exam.
View step feedback	Student	/student/review/<pk>/feedback/<answer_id>/	Step-by-step explanation after session.
AI tutor	Student	/student/review/<pk>/tutor/chat/	Post-session AI assistance when provider configured.
Session history	Student	/student/sessions/	Past review sessions.
View mistakes	Student	/student/mistakes/	Auto-logged wrong answers.
Weak areas	Student	/student/mistakes/patterns/	Mistake patterns by topic.
Topic progress	Student	/student/topics/	Mastery progress per topic.

2.6 Faculty — Analytics and Intervention

Function	Who	System Location (URL)	Logic
Cross-course overview	Faculty	/professor/overview/	Summary across all owned courses.
Course analytics	Faculty	/professor/courses/<pk>/	Calibration breakdown, intervention flags.
Course roster	Faculty	/professor/courses/<pk>/roster/	Students who participated in review for this course.
Student drill-down	Faculty	/professor/courses/<course_pk>/roster/<student_pk>/	Individual student performance in course context.
Topic heatmap	Faculty	/professor/courses/<pk>/heatmap/	Visual topic-level performance.
Export interventions CSV	Faculty	/professor/courses/<pk>/interventions/export/	Flagged students with accuracy and suggested actions.

Part 3 — Step-by-Step User Flows (Pure Logic)

Flow A — Account Lifecycle

1. User visits landing page (/) or goes directly to /accounts/signup/.
2. User selects role: Student, Faculty, or Chairperson and submits registration form.
3. If Student: account is approved and active immediately; redirect to /student/dashboard/.
4. If Faculty or Chairperson: account is pending and inactive; user sees pending message at login.
5. Campus administrator opens /campus/users/, reviews pending accounts, approves or rejects.
6. On approval: faculty redirect to /professor/courses/; chairperson to /chairperson/dashboard/.
7. All users maintain profile at /profile/ throughout their use of the system.

Flow B — Institutional Setup (Prerequisite Pipeline)

1. Campus admin sets current academic year and term at /campus/academic-calendar/.
2. Campus admin creates program sections at /campus/sections/ with capacity limits.
3. Chairperson creates teaching assignments at /chairperson/assignments/create/.
4. System auto-creates Course offering and ExamSetup for each assignment.
5. Faculty opens /professor/courses/<pk>/exam-setup/ and enables exam with topics, difficulties, timer.
6. Faculty creates questions at /professor/courses/<pk>/questions/; each enters PENDING status.
7. Chairperson approves questions (intended UI at question review; service layer enforces department scope).
8. Approved and active questions enter the student review pool.

Flow C — Student Review Session

1. Student sets home degree program, year level, and section in /profile/.
2. System checks eligibility: profile complete, year matches section, current term active, enabled exam setup exists for section.
3. If ineligible: block session; show eligibility message.
4. Student opens /student/review/setup/ (sidebar: Start Review).
5. Student selects subject, topic, difficulty, and duration via setup wizard.
6. System creates ReviewSession in timed_exam mode; redirect to /student/review/<pk>/.
7. Questions delivered adaptively (prior mistakes, high-confidence wrong answers, unseen items).
8. Per-question timer runs; confidence inferred from response time vs. time limit.
9. Student submits answers; system grades, logs mistakes on wrong answers.
10. Explanations are withheld during exam; brief answer reveal only in timed mode.
11. Session ends; redirect to /student/review/<pk>/summary/ with calibration insights.
12. Student may follow up via Session History, Mistakes, Weak Areas, Topic Progress.

Flow D — Faculty Instructional Response

1. Student session data aggregates into course analytics.
2. Faculty opens /professor/courses/<pk>/ or roster view.
3. System flags students on intervention list based on accuracy, calibration gaps, practice frequency.
4. Faculty drills into individual student at roster student detail page.
5. Faculty exports intervention CSV for follow-up at interventions export URL.
6. Faculty edits explanations on high-mistake questions at feedback list.
7. Optional: faculty generates AI course summary for reporting.

Flow E — Chairperson Institutional Oversight

1. Chairperson views /chairperson/dashboard/ for department-wide program metrics.
2. Chairperson manages teaching assignments to link faculty to sections and subjects.
3. Chairperson reviews pending questions and approves or rejects (department-scoped programs only).
4. Chairperson monitors cross-program analytics by student home degree program.
5. Chairperson audits course participation at /chairperson/courses/.

6. For thesis: chairperson exports anonymized research CSV with calibration and survey data.

Flow F — Pilot Study Participant (Thesis)

1. Student logs in and sets home degree program in /profile/.
2. Optional: submit pilot study consent (checkbox).
3. Complete pre-survey (Likert 1–5) from profile before or during pilot.
4. Student completes review sessions during pilot period (Flow C).
5. Submit post-survey from profile after using the system (no session gate).
6. Chairperson exports anonymized data: student hash, program, sessions, calibration counts, survey scores.

3.1 Question Lifecycle Logic

- Faculty creates or edits question → status PENDING.
- Chairperson approves → status APPROVED; question is active in sessions if is_active=True.
- Chairperson rejects → status REJECTED; not available in sessions.
- Faculty edits approved question → returns to PENDING; requires re-approval.
- Faculty deactivates question → removed from active pool regardless of status.

3.2 Calibration Classification Logic

Classification	Condition	Instructional Meaning
Mastery	High confidence + correct	Student knows the material well.
Misconception	High confidence + wrong	Confident but incorrect; priority intervention target.
Lucky guess	Low confidence + correct	Correct by chance; knowledge may be fragile.
Expected gap	Low confidence + wrong	Student aware of knowledge gap.
Uncertain	Medium confidence	Ambiguous calibration signal.

In timed exam mode, confidence is inferred from answer time relative to the per-question time limit (default 30 seconds, configurable in exam setup). Manual confidence ratings apply in practice review mode but student setup currently defaults to timed exam.

Part 4 — Sidebar Navigation by Role

Role	Primary Navigation Items
Student	Dashboard Start Review Session History Mistakes Weak Areas Topic Progress Profile
Faculty	Overview My Courses (Analytics, Masterlist, Exam Setup, Heatmap, Topics, Question Bank, Feedback)
Chairperson	Programs Assignments Faculty Logs Student Logs Cross-Program Course Audit
Campus Admin	Campus Dashboard Users Sections Academic Calendar

4.1 Eligibility Gate Logic (Student Review)

- Profile must have section and year level set.
- Year level must match assigned section.
- A current academic term must be active (campus admin responsibility).
- At least one teaching assignment for the student's section must have exam_setup.is_enabled=True.
- Approved, active questions must exist for selected topic and difficulty.

Part 5 — Panelist Questions and Answers

5.1 Problem Definition and Research Rationale

Q: What specific problem in self-regulated learning does ExamiQ+ address?

A: ExamiQ+ addresses the gap between what students think they know and what they actually know — called poor calibration. Many students are overconfident on topics they get wrong (misconceptions) or underconfident on topics they get right (lucky guesses). The platform makes this visible through timed review sessions, confidence inference, and a calibration summary so students can monitor and adjust their study behavior — the core of self-regulated learning.

Q: Why focus on calibration awareness rather than raw exam scores?

A: Raw scores only show whether an answer was right or wrong. Calibration shows whether the student knew why they were right or wrong. A student with 70% accuracy but many misconceptions is in a different learning state than one with 70% and mostly expected gaps. Calibration awareness is the metacognitive skill that helps students decide what to review next, which is the thesis outcome measured by pre- and post-surveys.

Q: How is your study different from existing LMS quiz modules or commercial review apps?

A: Standard LMS quizzes report scores only. Commercial apps often give immediate explanations during practice, which reduces exam-like pressure. ExamiQ+ combines timed exam conditions (no inline explanations), inferred confidence from response time, a five-tier calibration matrix (mastery, misconception, lucky guess, expected gap, uncertain), role-based analytics for faculty intervention, and department-scoped content governance with chairperson question approval.

Q: Why mathematics only, and why scoped by degree program instead of a shared question pool?

A: Mathematics is a common required subject across all eight degree programs, making cross-program comparison possible within one campus. Each program has different math needs (e.g., discrete math for CS vs. business math for Marketing), so content is scoped by home degree program rather than a single shared pool. This ensures review questions match the student's actual curriculum.

Q: What gap in the literature does your thesis fill?

A: Existing SRL research often relies on self-reported confidence ratings, which interrupt exam flow and may be unreliable under time pressure. This study evaluates whether a web platform that infers confidence from response time, surfaces calibration gaps after timed sessions, and provides faculty intervention tools can improve students' calibration awareness as measured by validated survey items before and after use.

Q: Who are your target participants and why are they appropriate for this study?

A: College students enrolled in one of the eight supported degree programs who are actively taking math subjects during the pilot term. They are appropriate because they face real exam pressure, have program-specific math content available in the system, and can complete multiple review sessions within a single academic term to show measurable change in calibration awareness.

5.2 Theoretical Framework

Q: What theories of self-regulated learning underpin your design?

A: The design draws on Zimmerman's cyclical model of SRL (forethought, performance, self-reflection) and Nelson and Narens' metacognitive monitoring framework. Students plan review (forethought via topic and difficulty selection), monitor performance under timed conditions, and reflect on calibration results on the session summary. Faculty analytics support the social/contextual dimension of SRL through instructional intervention.

Q: How do you define calibration awareness in operational terms?

A: Calibration awareness is the student's ability to recognize the relationship between their confidence and their actual performance. Operationally, it is measured by the pre- and post-survey Likert item: 'I am aware of how confidence affects learning' (pre) and 'EXAMIQ helped me understand my calibration gaps' (post). In the system, it is represented by the calibration matrix counts per session and aggregated on the student dashboard.

Q: What is the relationship between confidence and metacognition in your model?

A: Confidence is treated as a behavioral proxy for metacognitive judgment — the student's implicit estimate of knowing. Fast answers suggest high perceived certainty; slow answers suggest doubt. When this proxy is crossed with correctness, the system classifies the metacognitive state (well-calibrated mastery vs. miscalibrated misconception). The session summary makes this explicit so students can engage in metacognitive reflection.

Q: Which SRL phases (planning, monitoring, control) does ExamiQ+ support?

A: Planning: review setup wizard (subject, topic, difficulty, duration). Monitoring: per-question timer and real-time answer submission. Control: session summary recommendations, mistake list, weak-area patterns, and topic progress guide what to study next. Faculty intervention lists extend control to the instructional level.

Q: How do your calibration tiers (mastery, misconception, lucky guess, expected gap) map to theory?

A: These map to the confidence × accuracy matrix used in metacognitive research. Mastery = high confidence + correct (accurate monitoring). Misconception = high confidence + wrong (overconfidence). Lucky guess = low confidence + correct (underconfidence). Expected gap = low confidence + wrong (accurate awareness of not knowing). Uncertain = medium confidence, ambiguous signal requiring further practice.

Q: What prior research supports inferring confidence from response time instead of self-report?

A: Cognitive psychology literature links response latency to retrieval strength and judgment of learning. Fast correct answers typically indicate fluent recall; slow incorrect answers suggest struggle. ExamiQ+ applies fixed thresholds scaled to the per-question time limit: ≤33% of limit = high, ≤73% = medium, above = low. This avoids interrupting timed exams with extra rating steps while still producing a confidence signal for calibration analysis.

5.3 System Design and Architecture

Q: Why did you choose Django and this role-based architecture?

A: Django provides built-in authentication, ORM, admin, and security features suitable for a multi-role academic system. The role-based architecture (student, professor, chairperson, campus admin) mirrors real institutional hierarchy. Each role has its own URL namespace (/student/, /professor/, /chairperson/, /campus/) enforced by mixins that return HTTP 403 on unauthorized access.

Q: How do you enforce separation of duties between faculty, chairperson, and campus admin?

A: Campus admin approves accounts and manages calendar/sections. Chairperson creates teaching assignments and approves questions for programs their department manages. Faculty configure exams and create questions but cannot approve their own submissions. Students can only review within their home program. Each boundary is enforced in view mixins (StudentRequiredMixin, ProfessorCourseMixin, ChairpersonProgramMixin, CampusAdminRequiredMixin).

Q: Explain the data flow from question creation to student review to analytics.

A: Faculty creates a question → status PENDING. Chairperson approves → APPROVED. Student starts review → system selects approved active questions adaptively. Each answer is graded, confidence inferred, and stored. Wrong answers create MistakeRecord entries. Session completion feeds student dashboard, faculty course analytics, chairperson program dashboard, and research export.

Q: How does program scoping prevent students from accessing other departments' content?

A: Each student has a home_degree_program field on their profile. Review setup filters subjects and topics to that program's curriculum. Course offerings, question banks, and analytics are linked to program slugs. Querysets in student views filter by the logged-in student's program; cross-program URLs are not exposed in the student interface.

Q: What is the adaptive question selection algorithm and what pedagogical goal does it serve?

A: Questions are weighted-randomly selected: +3 weight if the topic has prior mistakes, +2 if the specific question was previously answered wrong with high confidence, +1 if unseen. A 10% chance of pure random selection prevents predictability. The goal is spaced retrieval practice focused on weak areas and misconceptions rather than repeating already-mastered items.

Q: Why are explanations shown only on the session summary and not during the timed exam?

A: To simulate exam conditions where students cannot look up answers mid-test. This preserves the integrity of the timed confidence signal — immediate explanations would change both performance and response time. Post-session explanations on the summary support the reflection phase of SRL without contaminating the monitoring phase.

Q: How does the system handle concurrent users during a pilot?

A: Each review session is independent per student. Django serves stateless HTTP requests; session state is stored in PostgreSQL. Question delivery uses per-session exclude lists to avoid repeats. For pilot scale (dozens to low hundreds of students), standard single-server deployment with database connection pooling is sufficient. No shared in-memory state is required between users.

5.4 User Roles and Workflows

Q: Walk us through what a student does from registration to session summary.

A: 1) Register at /accounts/signup/ with program, year level, section. 2) Land on /student/dashboard/. 3) Complete profile if needed. 4) Click Start Review → /student/review/setup/. 5) Select subject, topic, difficulty, duration. 6) Take timed exam at /student/review/<pk>/ answering each question within the countdown. 7) View session summary at /student/review/<pk>/summary/ showing accuracy, calibration matrix, and recommended topics. 8) Optionally review mistakes and topic progress.

Q: What must happen institutionally before a student can start a review session?

A: Campus admin must set the current academic term and create sections. Chairperson must create a teaching assignment linking faculty, section, subject, and term (auto-creates course and exam setup). Faculty must enable exam setup with topics and timer settings. Faculty must create questions; chairperson must approve them. Student profile must have matching program, year level, and section.

Q: Why do faculty and chairperson accounts require campus approval but students do not?

A: Faculty and chairperson accounts grant access to student data, content management, and institutional analytics. Requiring campus admin approval verifies employment and prevents unauthorized persons from accessing sensitive educational records. Student accounts only access their own data and publicly available review content within their program.

Q: What is the chairperson's role in question quality control?

A: Chairpersons review faculty-submitted questions and approve or reject them before they enter the student review pool. Approval is limited to programs managed by the chairperson's department. This ensures mathematical accuracy, appropriate difficulty, and alignment with program curriculum. The approval service layer exists in the codebase; the dedicated review UI is planned.

Q: How does a professor identify and intervene with at-risk students?

A: Faculty opens course analytics at /professor/courses/<pk>/ which shows an intervention list. Students are flagged based on low accuracy, high misconception count, low practice frequency (no sessions in 14 days), and calibration gaps. Faculty drills into individual students via the roster, then exports an intervention CSV with suggested actions for follow-up.

Q: What can the campus administrator do that other roles cannot?

A: Approve or reject faculty and chairperson registrations, directly provision staff accounts, manage program sections and capacity, set the current academic year and term, and access the campus dashboard. No other role can activate pending staff accounts or control the academic calendar that gates student review eligibility.

5.5 Confidence and Calibration Measurement

Q: How is confidence measured in timed exam mode?

A: Confidence is not manually rated during the exam. The system records time spent per question and maps it to a confidence tier (1 = low, 3 = medium, 5 = high) using the confidence_from_time_spent function in apps/analytics/confidence.py. This value is stored on each Answer record alongside correctness for calibration classification.

Q: What are the thresholds for high, medium, and low confidence from response time?

A: Thresholds scale to the per-question time limit (default 30 seconds). At 30s limit: $\leq 10s$ = high confidence (5), $\leq 22s$ = medium (3), $> 22s$ = low (1). For other limits, thresholds scale proportionally (33% and 73% of the limit). These constants are HIGH_CONFIDENCE_MAX_SECONDS=10 and AVERAGE_CONFIDENCE_MAX_SECONDS=22 relative to CONFIDENCE_TIMER_CAP_SECONDS=30.

Q: Why omit manual confidence ratings during the timed exam?

A: Manual ratings after each question add cognitive load and break exam flow, potentially changing both response times and answer accuracy. Omitting them preserves a natural timed-exam experience while still producing a confidence proxy. The pilot protocol explicitly states confidence ratings are omitted in timed exam mode; explanations appear on the summary instead.

Q: How do you validate that inferred confidence relates to actual student certainty?

A: Validation is done through convergent evidence: (1) post-survey item on confidence rating usefulness, (2) correlation between response time distributions and self-reported calibration awareness change, (3) face validity — fast wrong answers (misconceptions) match the intuitive overconfidence pattern. Full psychometric validation of time-based inference is acknowledged as a limitation and future work.

Q: What is a misconception in your system and why is it flagged for intervention?

A: A misconception is a high-confidence wrong answer — the student answered quickly (suggesting certainty) but was incorrect. This is the most dangerous calibration state because the student believes they understand material they do not. The intervention list prioritizes students with high misconception counts, and the adaptive algorithm re-serves those questions with extra weight.

Q: Can a student game the system by answering quickly or slowly to manipulate confidence?

A: Partially. A student could deliberately slow down to lower confidence scores, but this also risks timer expiry and wrong answers. The timed constraint limits gaming — slowing down reduces time for thinking. Confidence is one input among many (accuracy, mistake patterns, session frequency) for analytics and research. Gaming would hurt their own performance metrics.

5.6 Pilot Study Methodology

Q: What is your research design: pre-experimental, quasi-experimental, or other?

A: A one-group pre-test/post-test pre-experimental design. Each participant completes a pre-survey before using ExamiQ+ and a post-survey after the pilot period. Change in calibration awareness and related Likert items is measured within subjects. There is no control group in the initial campus pilot; this is appropriate for a system feasibility and preliminary effectiveness study.

Q: How many participants do you need and how did you compute sample size?

A: The pilot targets students across participating programs within one academic term. Sample size is determined by available enrolled students who consent to participate. For a paired pre-post Likert study, a minimum of 30 participants is commonly cited for central limit theorem applicability in thesis pilots. The department-scoped export allows aggregation across programs to reach this target.

Q: What are your pre- and post-survey instruments and are they validated?

A: Three Likert items (1–5) per survey: calibration awareness, confidence rating usefulness, and would recommend, plus optional open feedback. Pre-survey asks about general awareness; post-survey asks about ExamiQ-specific impact. Items are adapted from SRL and calibration literature. Formal validation with pilot reliability analysis (Cronbach's alpha) is planned as part of thesis analysis.

Q: What are your dependent and independent variables?

A: Dependent variables: post-survey scores (calibration awareness, confidence usefulness, recommendation), change scores (post minus pre), and behavioral metrics (misconception count, accuracy, sessions completed). Independent variable: exposure to ExamiQ+ (usage duration and number of review sessions). Covariates: home degree program, year level, baseline pre-survey scores.

Q: How long is the pilot period and how many sessions must each student complete?

A: The pilot runs for one academic term (duration set by campus admin via academic calendar). Students are encouraged to complete multiple review sessions but there is no hard session gate for the post-survey — it can be submitted anytime from Profile after using the system. More sessions produce richer calibration data for analysis.

Q: How do you control for prior math ability or study habits?

A: Pre-survey scores serve as a baseline for within-subject comparison. Program and year level are recorded as covariates in the research export. The system logs session frequency to distinguish active from inactive participants. Full control of prior ability would require a standardized math pre-test, which is noted as a limitation of the current design.

Q: What ethical approvals and consent procedures are in place?

A: Participants submit optional pilot consent via a checkbox (PilotConsent model). Pre- and post-surveys are voluntary. Research export uses SHA-256 hashed student IDs with no names or emails. Institutional ethics review board approval is obtained per university requirements before data collection begins.

5.7 Data Collection and Analysis

Q: What data does the system log for research purposes?

A: Per student: sessions completed, total answers, accuracy percentage, average confidence, counts per calibration tier (mastery, misconception, lucky guess, expected gap, uncertain), mistake records, home program, and pre/post survey Likert scores. All exported via the chairperson research CSV endpoint.

Q: How is student identity protected in the research export?

A: The export uses student_hash — the first 16 characters of SHA-256 hash of the student ID — with no name, email, or student number. Data is department-scoped so only the chairperson of the managing department can export. Faculty intervention CSVs contain names but are separate from the anonymized research export.

Q: What statistical tests will you use to compare pre- and post-survey scores?

A: Paired t-test or Wilcoxon signed-rank test (if normality assumption fails) for each Likert item comparing pre vs. post scores. Effect size (Cohen's d) will be reported. Descriptive statistics (mean, SD, frequency) for calibration tier distributions. Correlation analysis between session count and calibration awareness change.

Q: How will you analyze calibration tier distributions before and after intervention?

A: Early-session calibration profiles (first N sessions) are compared to later-session profiles for the same student using misconception rate and mastery rate. A reduction in misconception proportion and increase in mastery proportion over time indicates improved calibration. Aggregated tier counts from the research export support group-level analysis.

Q: What constitutes a meaningful improvement in calibration awareness?

A: A statistically significant increase in the post-survey calibration awareness item ($p < 0.05$) with at least a medium effect size (Cohen's $d \geq 0.5$). Practically, a mean increase of ≥ 0.5 points on the 5-point Likert scale is considered educationally meaningful for a short pilot.

Q: How do you handle incomplete sessions or students who drop out of the pilot?

A: Partial sessions are saved with their completed answers and appear in session history. Dropouts are included in analysis only if they completed the pre-survey (intent-to-treat) or excluded in a per-protocol sensitivity analysis. Session count and completion status are recorded in the export to filter active participants.

5.8 AI Features

Q: What AI features are integrated and what is their role in the study?

A: AI question generation (faculty drafts questions from topic prompts), AI difficulty suggestion, adaptive step-by-step feedback generation, AI tutor chat post-session, and optional AI course summary for faculty. AI is a supporting tool, not the primary intervention. The core thesis intervention is the calibration-aware review workflow itself.

Q: How does AI question generation maintain quality and alignment with learning objectives?

A: AI-generated questions enter the same PENDING approval workflow as manual questions. Faculty review and edit drafts before submission. Chairperson must approve before questions reach students. This human-in-the-loop gate ensures curriculum alignment regardless of AI output quality.

Q: Does the AI tutor influence SRL outcomes and how do you account for that in analysis?

A: The AI tutor is optional and only available post-session. Usage is not required for the pilot. If tutor usage data is collected, it can be treated as a covariate. The primary SRL outcome is calibration awareness from surveys and behavioral calibration metrics, which are independent of tutor interaction.

Q: What LLM provider do you use and how do you handle API failures or hallucinations?

A: The system supports configurable LLM providers (e.g., Gemini when LLM_PROVIDER=gemini and API key is set). API failures return graceful error messages without blocking core review functionality. AI-generated content is always reviewed by faculty before student exposure. Hallucinated math content is caught by the approval workflow.

Q: Are AI-generated explanations reviewed by faculty before students see them?

A: Faculty can edit explanations at /professor/courses/<pk>/feedback/. AI-generated feedback is a draft that faculty can modify. Questions must be chairperson-approved before entering sessions. This two-layer review (faculty edit + chair approval) limits exposure to unreviewed AI content.

5.9 Limitations and Scope

Q: What are the limitations of a single-campus pilot?

A: Results may not generalize to other institutions with different curricula, student populations, or technology access. Sample size is constrained to enrolled students in participating programs. No control group limits causal claims. External validity is addressed by documenting the implementation precisely so others can replicate.

Q: Is the system generalizable beyond the eight degree programs?

A: The architecture is program-agnostic — adding a program requires seeding subjects, topics, and a managing department. The eight current programs demonstrate the pattern. Generalization to non-math subjects would require new question types and grading logic but the role and analytics framework would remain the same.

Q: What features are planned but not yet fully implemented?

A: Chairperson question review UI (service exists, URL not wired), research export URL registration, pilot consent and survey forms on the Profile page (models exist), practice review mode in student setup (model supports it but setup defaults to timed_exam), and AI question variation confirm-save flow.

Q: How does timed exam mode differ from practice review mode and which did you use in the pilot?

A: Timed exam: per-question countdown, confidence inferred from time, no inline explanations. Practice review: supports manual confidence ratings and a more relaxed pace. The pilot uses timed exam mode exclusively, as stated in the pilot protocol, to simulate exam conditions.

Q: What happens if no chairperson approves questions in time for the pilot?

A: Questions remain in PENDING status and are excluded from the review pool. Mitigation: pre-seed approved questions before the pilot, use Django admin for bulk approval, or run the auto-approve migration in development. Faculty are notified that submissions require approval.

Q: Can results be attributed to the platform versus normal classroom instruction?

A: Not with certainty in a one-group design. Classroom instruction continues during the pilot. The pre-post survey isolates self-reported calibration awareness change attributable to platform use. Acknowledging this confound is part of the limitations section. A future RCT with a control group would strengthen causal claims.

5.10 Security, Privacy, and Ethics

Q: How is student data stored and who can access it?

A: All data is stored in PostgreSQL. Students access only their own sessions, answers, and profile. Faculty access students enrolled in their courses via roster and analytics. Chairpersons access department-scoped aggregated data. Campus admins manage accounts. Django's ORM and view mixins enforce queryset filtering per role.

Q: What personal identifiable information is excluded from research exports?

A: The research CSV excludes name, email, student number, and phone. Only student_hash (SHA-256 truncated), program slug, and behavioral metrics are included. Survey scores are linked to the hash, not to identifiable fields.

Q: How do you comply with data privacy regulations applicable to your institution?

A: Consent is obtained before data collection. Data minimization is applied in research exports. Access is role-restricted. Passwords are hashed by Django's auth system. The platform follows the institution's data privacy policy and the Philippines Data Privacy Act requirements as applicable to educational records.

Q: Can faculty see individual student calibration data and is that ethically appropriate?

A: Yes — faculty see per-student accuracy, misconception counts, and calibration breakdowns for students in their courses. This is ethically appropriate because faculty already have a legitimate educational interest in student performance, the data supports instructional intervention (not punishment), and students are informed during pilot consent.

Q: How are passwords and authentication handled?

A: django-allauth with email-based login (USERNAME_FIELD = email). Passwords are hashed using Django's PBKDF2 hasher. CSRF protection on all forms. Session cookies for authentication. Pending/rejected accounts are blocked at the adapter level before login succeeds.

5.11 Testing and Validation

Q: How did you functionally test each role's workflows?

A: Documented in docs/functional-testing.md with Satisfactory/Unsatisfactory checklist per role (Appendix B) and detailed test cases (Appendix A). Each major function — signup, review session, exam setup, question creation, analytics — was tested against expected system responses. Cross-role access was verified (Appendix C): students receive 403 on faculty URLs, etc.

Q: What test cases cover eligibility gates and cross-role access control?

A: Student review setup blocks when profile is incomplete, section is mismatched, or no current term exists. Cross-role tests verify 403 for students on /professor/, faculty on /student/dashboard/, chairperson on /campus/, and anonymous redirect to login. These are in Appendix A and C of the functional testing document.

Q: Did you conduct usability testing with actual students?

A: The pilot itself serves as the primary usability evaluation with real students. Pre-pilot, demo accounts (e.g., student3@examiq.edu with misconception profile) were used for walkthrough testing. Post-survey open feedback captures usability impressions. Formal think-aloud usability testing is recommended as future work.

Q: How do you know the grading logic is correct for all question types?

A: The grade_answer function in apps/questions/services.py handles multiple choice, true/false, numeric, and short-answer types with type-specific comparison logic. Unit tests in the questions and reviews apps verify grading outcomes. Faculty review of question correctness during the approval workflow provides an additional human validation layer.

Q: What is your plan if the system fails during a live pilot session?

A: Completed answers are saved per submission — a mid-session failure preserves partial progress. Students can start a new session. Faculty and campus admin monitor via Django admin. A backup database and documented restart procedure are maintained. Health check endpoint at /health/ supports uptime monitoring.

5.12 Demonstration and Practical Scenarios

Q: Can you demonstrate a student completing a review and show the calibration summary?

A: Yes. Log in as student3@examiq.edu (demo1234). Go to Start Review, complete a session, then view /student/review/<pk>/summary/. The summary shows accuracy percentage, calibration matrix with mastery/misconception/lucky guess/expected gap counts, and a recommended topic for further study.

Q: Show how a professor uses the intervention list to identify a misconception profile.

A: Log in as `prof.algebra@examiq.edu` (demo1234). Open MATH101-CS course at `/professor/courses/<pk>/`. The intervention list flags student3 with high misconception count. Click into the student detail from the roster to see per-answer calibration breakdown. Export CSV for advising.

Q: Show how a chairperson views department-wide program performance.

A: Log in as `chair.computersci@examiq.edu` (demo1234). Open `/chairperson/dashboard/` for per-program accuracy and session counts within the CS department scope. Cross-program view at `/chairperson/analytics/by-program/` compares all programs managed by the department.

Q: What does the research CSV export contain and how would you use it in analysis?

A: Columns: `student_hash`, `program`, `sessions_completed`, `total_answers`, `accuracy_pct`, `avg_confidence`, five calibration tier counts, `mistake_records`, and pre/post survey Likert scores. Import into SPSS or Python pandas for paired t-tests on survey items, descriptive stats on calibration distributions, and correlation between usage intensity and awareness change.

Q: What would you show if the panel asks about a student who is blocked from starting review?

A: Navigate to `/student/review/setup/` with an incomplete profile. The system displays the eligibility message listing missing requirements: no section, no current term, or no enabled exam setup. Demonstrate fixing each condition — set profile fields, campus admin sets term, faculty enables exam setup — and show the wizard proceeding.

5.13 Future Work and Sustainability

Q: What improvements would you make after the pilot?

A: Wire the chairperson question review UI and research export URL, add pilot consent and survey forms to Profile, enable practice review mode as a student option, validate time-based confidence against self-report in a follow-up study, and add a control group for stronger causal evidence.

Q: Can the platform scale to more programs or institutions?

A: Yes. The program model is data-driven — new programs are added via fixtures or admin without code changes. Multi-tenancy (separate campuses) would require an institution field on core models. Current single-campus design scales to thousands of students on standard Django deployment with PostgreSQL.

Q: Will the system remain deployed after the thesis defense?

A: The platform is designed for continued institutional use beyond the thesis. Campus admin and chairperson workflows support ongoing term-over-term operation. Pilot data collection can be disabled while retaining review and analytics functionality for regular academic use.

Q: How would you add support for non-math subjects?

A: Add new subjects and topics under existing or new programs. Extend the `grade_answer` function for new question types (essay rubric, code evaluation). AI feedback prompts would need subject-specific templates. The role architecture, session flow, and analytics framework are subject-agnostic.

Q: What metrics would you track for long-term institutional adoption?

A: Active students per term, sessions per student per week, question bank growth rate, approval turnaround time, faculty intervention CSV downloads, misconception rate trends across terms, and student post-survey recommendation scores. Chairperson dashboard metrics provide institutional-level adoption visibility.

Appendix — Function-to-Location Quick Reference

I want to...	Go to...
Register	/accounts/signup/
Log in	/accounts/login/
Update my profile	/profile/
Approve a faculty account	/campus/users/
Set the current semester	/campus/academic-calendar/
Create student sections	/campus/sections/
Assign faculty to a class	/chairperson/assignments/create/
Enable exams for a course	/professor/courses/<pk>/exam-setup/
Add questions	/professor/courses/<pk>/questions/
Approve questions	Chairperson question review (intended URL)
Start a review session	/student/review/setup/
See my calibration results	/student/review/<pk>/summary/
See my mistakes	/student/mistakes/
See class performance	/professor/courses/<pk>/
Export intervention list	/professor/courses/<pk>/interventions/export/
See department analytics	/chairperson/dashboard/
Export thesis research data	/chairperson/research/export/ (intended URL)